
Training a 2-token attention is NP-hard

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Abstract

As the title indicates, this work is inspired by the celebrated paper of Blum and Rivest, “Training a 3-node neural network is NP-complete.” They showed that for a 2-layer, 3-node, n -input neural network (arguably the simplest neural network), deciding whether weights exist that fit a given training set exactly is NP-complete. We show that for an attention head with 1-dimensional query, key, and value, and n -dimensional token input (arguably the simplest attention head), the analogous exact-fit problem on a training set of 2-token sequences is NP-hard; and that if softmax is replaced with hardmax, the decision problem becomes NP-complete.

1 Introduction

One of the best-known classical results in neural networks is that of Blum and Rivest [1], establishing that training even a 3-node neural network is an NP-complete problem. The staying power of this result is impressive: Nearly four decades later, and limiting to this conference series alone, we see follow-up works nearly every year [2–4]. It is therefore somewhat surprising that the result of Blum and Rivest has not been extended from neural networks to their modern counterpart, the attention head [5].

A surprising aspect of Blum and Rivest’s intractability result is the simplicity of the neural network they considered, just 2-layer, 3-node. Naturally we expect training attention to be harder than training neural networks but to what extent? We show that even when the dimensions of query, key, and value are lowered to the point of near-triviality, namely just 1-dimensional, i.e., the attention module associates each token with a scalar-valued key, query, and value (as opposed to vector-valued ones); and when training examples are just 2-token, again the lowest possible nontrivial value, the problem of training an attention head is already NP-hard.

1.1 Related work

As we have alluded to, there has been an abundance of work following [1] on the computational complexity of neural-network training and learning, including hardness results for small mean-squared-error objectives [6], approximation problems and richer architectures [7–9], fixed input dimension and growing number of hidden neurons [10], other complexity classes like $\exists\mathbb{R}$ -completeness [11], and positive algorithmic results for special ReLU settings [12]. Their common goal is to sharpen the contrast between the apparent simplicity of neural network architectures and the worst-case complexity of fitting their parameters.

To the best of our knowledge, there has been no similar investigation for the attention head. For sure there have been many studies of the expressivity, language-recognition power, and computational limits of transformer architectures [13–15], typically asking which formal languages or function classes can be represented or computed by transformer-style models. But when it comes to complexity theoretic results, the closest related work we could find is Likhoshesterov et al. [16], which shows that for fixed key dimension greater than one, deciding whether a target self-attention matrix can be weakly approximated is NP-hard by jointly choosing the input X and parameters W_Q, W_K . They also prove

separate positive approximation guarantees for sparse target matrices. Other differences aside, their setting departs from the minimalist spirit of Blum and Rivest’s result in ?.

Our work on the other hand strictly adheres to this minimalism, considering only a single attention head with just real-valued (not vector-valued) parameters, and trained with mere two-token examples — any lower and we lose the attention mechanism (but just for argument sake, we did look at what happens with “one-token” examples in Proposition ??). Like Blum and Rivest ?, and in contrast to ?, we fix the input X as part of the training data, and seek to learn the weights W_Q, W_K, W_V in exact zero-loss decision problems. In particular, we do not seek an approximation as in ? but require an exact fit of the attention output as Blum and Rivest did with neural networks in ?.

So as to leave no ambiguity, neither the hardness result for attention in ? nor those for neural networks in any of the aforementioned references would imply the NP-hardness of exact fitting for a single attention head (Theorem ??) and other hardness results (Theorems ?? and ??) claimed in this article.

1.2 NP-hard versus NP-complete

A small deviation from ? is that we prove NP-hardness instead of NP-completeness. The main reason is that attention as defined in ? requires working with softmax and the Cook–Karp–Levin notion of NP-completeness requires working in the Turing bit model. These are fundamentally incompatible: Turing bit model cannot easily accommodate a discussion of transcendental functions.¹ Blum and Rivest faced no such difficulty as their neural network uses the sign function as activation. Indeed, as soon as we replace softmax by its zero-temperature limit, the hardmax, the NP-completeness of a hardmax attention (Theorem ??) follows almost immediately.

2 Model and exact-fit decision problems

We first fix the attention notation, the two exact-fit decision problems, and the 1-in-3-SAT variant from which we reduce.

2.1 Single-head attention layer

Throughout the paper we adopt the column-major convention for inputs: each sample is a matrix $X \in \mathbb{R}^{n \times p}$ whose p columns are n -dimensional input vectors. In large language models, a column vector in X is usually called a token. A *single-head attention layer* ? with parameters

$$W_Q \in \mathbb{R}^{d_k \times n}, \quad W_K \in \mathbb{R}^{d_k \times n}, \quad W_V \in \mathbb{R}^{d_v \times n},$$

collectively denoted $\Theta = (W_Q, W_K, W_V)$, maps $X \in \mathbb{R}^{n \times p}$ to $\mathbb{R}^{d_v \times p}$ via

$$A_S(X; \Theta) = W_V X \operatorname{softmax}(X^\top W_Q^\top W_K X) \quad (1)$$

in the softmax case, and via

$$A_H(X; \Theta) = W_V X \operatorname{hardmax}(X^\top W_Q^\top W_K X) \quad (2)$$

in the hardmax case. $W_Q X, W_K X, W_V X$ compute the query, key, and value, respectively. Both softmax and hardmax are applied *column-wise* to the $p \times p$ score matrix $X^\top W_Q^\top W_K X$. For a vector $z \in \mathbb{R}^p$,

$$\operatorname{softmax}(z)_i = \frac{e^{z_i}}{e^{z_1} + \dots + e^{z_p}}, \quad \operatorname{hardmax}(z) = \lim_{\tau \rightarrow +\infty} \operatorname{softmax}(\tau z),$$

so hardmax assigns weight $1/k$ to each of the k argmax coordinates and 0 elsewhere, the tie-averaging rule called *average hard attention* in the formal-language transformer literature ??). (Alternatively, one can define a tie-breaking rule that assigns weight 1 to the first argmax coordinate and 0 elsewhere. It is not hard to see our argument still works for this tie-breaking hardmax *mutatis mutandis*.)

We write $M = W_Q^\top W_K \in \mathbb{R}^{n \times n}$ for the bilinear score matrix, and for two columns $a, b \in \mathbb{R}^n$,

$$s(a, b) = a^\top M b = (W_Q a)^\top (W_K b).$$

¹This is one reason optimization theory often uses oracle-complexity models to abstract away from bit-level representations; see ?.

We write $v(a) = W_v a \in \mathbb{R}^{d_v}$ for the value of a . Concretely, the j th output column of (??) is

$$A_s(X; \Theta)_{\bullet, j} = \sum_{i=1}^p v(x_i) \alpha_{i,j}, \quad \alpha_{\bullet, j} = \text{softmax}((s(x_i, x_j))_{i=1}^p), \quad (3)$$

and analogously for hardmax. Our hardness reductions construct a rank-one matrix $M = qk^\top$ for vectors $q, k \in \mathbb{R}^n$.

2.2 Exact-fit decision problems

Both decision problems below take a finite rational dataset and ask whether real parameters exist that fit it exactly; they differ only in the attention variant (softmax versus hardmax).

Problem (SOFTMAX-EXACT-FIT).

Input: A finite dataset $\mathcal{D} = \{(X_t, Y_t) : t = 1, \dots, N\}^2$ where $X_t \in \mathbb{Q}^{n \times p_t}$ and $Y_t \in \mathbb{Q}^{d_v \times p_t}$.

Output: Decide whether there exist real parameters $W_Q, W_K \in \mathbb{R}^{d_k \times n}$ and $W_V \in \mathbb{R}^{d_v \times n}$ such that $A_s(X_t; \Theta) = Y_t$ for every $t = 1, \dots, N$.

Problem (HARDMAX-EXACT-FIT).

Input: The same finite dataset as in SOFTMAX-EXACT-FIT.

Output: Decide whether there exist real parameters Θ such that $A_h(X_t; \Theta) = Y_t$ for every $t = 1, \dots, N$.

2.3 Length-one tractability

Before stating our hardness results we note that, with sequence length restricted to one, both decision problems collapse to rational linear feasibility.

Proposition 1. *The restrictions of SOFTMAX-EXACT-FIT and HARDMAX-EXACT-FIT to instances with $p_t = 1$ for every $t \in \{1, \dots, N\}$ are decidable in polynomial time.*

Proof. A length-one input has a single column, so both softmax and hardmax assign weight 1 and $A_s([x_t]; \Theta) = A_h([x_t]; \Theta) = W_v x_t$, independent of (W_Q, W_K) . The exact-fit conditions thus reduce to the rational linear system $W_v x_t = y_t$, decidable in polynomial time. $\square$

Thus the reductions below must involve length-two samples; the remaining preliminary is the restricted 1-in-3-SAT source problem.

2.4 Source problem: distinct-literal 1-in-3-SAT

Our reductions are from a slight restriction of the classical NP-complete 1-IN-3-SAT.

Problem (1-IN-3-SAT).

Input: A collection of clauses $C = \{c_1, \dots, c_m\}$ on Boolean variables $x_1, \dots, x_d$, where each clause $c_j = (\ell_{j,1}, \ell_{j,2}, \ell_{j,3})$ has three literals (each ℓ is some x_i or $\neg x_i$).

Output: Decide whether there is a truth assignment such that every clause has exactly one true literal.

The problem is NP-complete by Schaefer [Sch86]. The reductions of Sections ?? and ?? require the slight strengthening that clauses contain three *pairwise distinct* literals.

Lemma 1 (Distinct-literal restriction). *1-IN-3-SAT remains NP-complete when restricted to instances in which every clause has three pairwise distinct literals.*

²We index training samples by t throughout; the indices i and j are reserved for the SAT variable and clause indices in our reductions (Sections ??–??).

Proof. Given an instance ϕ , scan its clauses and rewrite each non-distinct clause as follows.

Case 1. A clause of the form (ℓ, ℓ, ℓ) (all three literals equal) is unsatisfiable in 1-in-3 semantics, so ϕ itself is unsatisfiable. Output two distinct-literal clauses

$$(x, y, \neg y), \quad (\neg x, z, \neg z)$$

on three new variables x, y, z . The first clause forces x to be false and the second forces x to be true, so the two new clauses are unsatisfiable.

Case 2. A clause of the form (ℓ, ℓ, ℓ') with $\ell \neq \ell'$ (after relabeling so the two equal literals are listed first) forces both copies of ℓ to take the same truth value; the 1-in-3 condition then forces ℓ to be false and ℓ' to be true. Replace this clause by the two distinct-literal clauses

$$(\ell, x, \neg x), \quad (\neg \ell', x, \neg x)$$

with a new variable x . Each new clause has three pairwise distinct literals. The two clauses force ℓ to be false and ℓ' to be true.

Applying these replacements independently to all non-distinct clauses gives an equisatisfiable instance of polynomial size, and membership in NP is immediate. $\square$

Sections ?? and ?? reduce from this distinct-literal restriction to the softmax and hardmax exact-fit problems respectively.

3 Softmax exact-fit is NP-hard

We begin with the softmax case, reducing from the distinct-literal restriction of 1-in-3-SAT (Lemma ??).

Theorem 1 (Softmax attention is NP-hard). *For every fixed $d_K \geq 1$ and $d_V \geq 1$, SOFTMAX-EXACT-FIT is NP-hard. Hardness already holds when every input sequence has length $p_t \leq 2$ and every entry of every X_t lies in $\{-2, -1, 0, 1\}$.*

We will first prove the theorem for the case $d_K = d_V = 1$. The general case is addressed in Section ??.

3.1 The reduction

Given a distinct-literal 1-IN-3-SAT instance ϕ with d variables $x_1, \dots, x_d$ and m clauses $c_j = (\ell_{j,1}, \ell_{j,2}, \ell_{j,3})$ (Lemma ??), set $n = 2d + 1$ and use the coordinate basis vectors

$$F, L_1, \dots, L_d, \bar{L}_1, \dots, \bar{L}_d \in \{0, 1\}^n.$$

For a literal ℓ , write $T(\ell) = L_i$ if $\ell = x_i$ and $T(\ell) = \bar{L}_i$ if $\ell = \neg x_i$. This means we will encode the literals x_i and $\neg x_i$ as vectors $L_i, \bar{L}_i$.

For each variable i and clause j , define the vectors

$$P_i = L_i + \bar{L}_i - F, \quad D_j = T(\ell_{j,1}) + T(\ell_{j,2}) + T(\ell_{j,3}) - 2F.$$

Because each clause has three pairwise distinct literals (Lemma ??), the three terms in D_j are distinct basis vectors, so D_j has coordinate 1 at three literal positions, -2 at F , and 0 elsewhere. Every input vector used in the reduction therefore has integer entries in $\{-2, -1, 0, 1\}$.

The dataset $\mathcal{D}_\phi$ consists of $4d + m + 1$ samples in six groups, with output dimension $d_V = 1$:

Group	Sample X_t	Target Y_t	Count
1	$[F]$	0	1
2	$[P_i]$	5	d
3	$[F, L_i]$	(1, 1)	d
4	$[F, \bar{L}_i]$	(1, 1)	d
5	$[F, P_i]$	(5/3, 5/2)	d
6	$[F, D_j]$	(7/3, 14/3)	m

The first-column equations for Groups 3–6 are easiest to read as equations in softmax odds, which we introduce next.

Define, for each variable x_i , the strictly positive scalars

$$\alpha_i = e^{s(F,F)-s(L_i,F)}, \quad \beta_i = e^{s(F,F)-s(\bar{L}_i,F)}, \quad \gamma_i = e^{s(F,F)-s(P_i,F)}, \quad (4)$$

which by bilinearity of s and $P_i = L_i + \bar{L}_i - F$ satisfy the exact identity $\gamma_i = \alpha_i \beta_i$. Groups 1–5 force $\alpha_i + \beta_i = 3$ and $\alpha_i \beta_i = 2$ (Lemma ??), pinning $\{\alpha_i, \beta_i\} = \{1, 2\}$ and yielding the Boolean assignment $x_i = \text{TRUE}$ iff $\alpha_i = 2$. Group 6 then forces exactly one true literal per clause (Lemma ??). All soundness arguments use only the first entry of Y_t ; the second entries are recovered automatically by the explicit completeness construction (Lemma ??).

3.2 Soundness

Soundness proceeds in two steps: the Variable Lemma extracts a Boolean assignment from the fitted parameters, and the Clause Lemma checks the 1-in-3 condition on each clause.

Lemma 2 (Variable Lemma). *If parameters Θ exactly fit Groups 1–5 of $\mathcal{D}_\phi$, then for every $i \in \{1, \dots, d\}$,*

$$\alpha_i + \beta_i = 3, \quad \alpha_i \beta_i = 2,$$

and consequently $\{\alpha_i, \beta_i\} = \{1, 2\}$ and $\{v(L_i), v(\bar{L}_i)\} = \{2, 3\}$.

Proof. The equation $A_s(X; \Theta) = Y$ for length-one samples $[F] \mapsto 0$ and $[P_i] \mapsto 5$ reduces to $v(F) = 0$ and $v(P_i) = 5$. For the length-two samples $[F, L_i]$, $[F, \bar{L}_i]$, $[F, P_i]$, the first entry of the same equation yields

$$\frac{v(L_i)}{1 + \alpha_i} = 1, \quad \frac{v(\bar{L}_i)}{1 + \beta_i} = 1, \quad \frac{v(P_i)}{1 + \gamma_i} = \frac{5}{3}.$$

Hence $v(L_i) = 1 + \alpha_i$, $v(\bar{L}_i) = 1 + \beta_i$, and $\gamma_i = 2$. Linearity of W_v together with $P_i = L_i + \bar{L}_i - F$ gives $v(L_i) + v(\bar{L}_i) = 5$, so $\alpha_i + \beta_i = 3$; bilinearity gives $\gamma_i = \alpha_i \beta_i$, hence $\alpha_i \beta_i = \gamma_i = 2$. The strictly positive scalars α_i, β_i are therefore the two roots of $z^2 - 3z + 2 = (z - 1)(z - 2)$, so $\{\alpha_i, \beta_i\} = \{1, 2\}$ and $\{v(L_i), v(\bar{L}_i)\} = \{2, 3\}$. $\square$

Define the Boolean assignment

$$x_i = \text{TRUE} \iff \alpha_i = 2 \iff v(L_i) = 3,$$

so a literal ℓ is true iff $\rho(\ell) = 2$, where $\rho(x_i) = \alpha_i$ and $\rho(\neg x_i) = \beta_i$.

Lemma 3 (Clause Lemma). *If Θ exactly fits Groups 1–6, then under the Boolean assignment above, every clause c_j has exactly one true literal.*

Proof. By bilinearity, with $\delta_j = e^{s(F,F)-s(D_j,F)}$,

$$s(F, F) - s(D_j, F) = \sum_{r=1}^3 (s(F, F) - s(T(\ell_{j,r}), F)), \quad \text{hence} \quad \delta_j = \prod_{r=1}^3 \rho(\ell_{j,r}).$$

Let k_j be the number of true literals in c_j . Then $\delta_j = 2^{k_j}$ and, since $v(T(\ell)) \in \{2, 3\}$ with $v = 3$ on true literals,

$$v(D_j) = \sum_{r=1}^3 v(T(\ell_{j,r})) - 2v(F) = 3k_j + 2(3 - k_j) = 6 + k_j.$$

The equation for the first entry of Y_t in Group 6 is therefore

$$\frac{v(D_j)}{1 + \delta_j} = \frac{6 + k_j}{1 + 2^{k_j}}.$$

Setting equal to $7/3$ leaves only $k_j = 1$:

k_j	0	1	2	3
$(6 + k_j)/(1 + 2^{k_j})$	3	7/3	8/5	1

$\square$

3.3 Completeness

Completeness is by explicit construction: we exhibit row-vector parameters (W_Q, W_K, W_V) derived from a satisfying assignment of ϕ and verify the targets group by group.

Lemma 4 (Completeness). *If ϕ is satisfiable, then there exist parameters Θ with $d_K = d_V = 1$ that exactly fits $\mathcal{D}_\phi$.*

Proof. Fix a satisfying assignment for ϕ . With $d_K = d_V = 1$ the parameters $W_Q, W_K, W_V \in \mathbb{R}^{1 \times n}$ are row vectors. Define them on the basis vectors by

Vectors	W_V	W_Q	W_K
F	0	0	1
True literal $T(\ell)$	3	$-\log 2$	1
False literal $T(\ell)$	2	0	0

extended linearly to all of $\mathbb{R}^n$. Then $v(F) = 0$ (Group 1) and, since exactly one of $L_i, \bar{L}_i$ is true, $v(P_i) = 3 + 2 = 5$ (Group 2). For Groups 3–6, the score matrix of $[F, U]$ has columnwise entries $(0, W_Q U)^\top$ and $(0, (W_Q U)(W_K U))^\top$ (using $W_Q F = 0, W_K F = 1$). The four resulting cases are:

Sample $[F, U]$ (case)	$(W_Q U, W_K U)$	F -col output	U -col output
$U \in \{L_i, \bar{L}_i\}, U$ true	$(-\log 2, 1)$	1	1
$U \in \{L_i, \bar{L}_i\}, U$ false	$(0, 0)$	1	1
$U = P_i$ (one literal true)	$(-\log 2, 0)$	$5/3$	$5/2$
$U = D_j$ (1-in-3 satisfied)	$(-\log 2, -1)$	$7/3$	$14/3$

For example, the D_j case has score matrix $\begin{pmatrix} 0 & 0 \\ -\log 2 & \log 2 \end{pmatrix}$, columnwise softmax $(2/3, 1/3)^\top$ and $(1/3, 2/3)^\top$, and $v(D_j) = 7$ (one true plus two false literals); the outputs are $(0, 7) \cdot (2/3, 1/3)^\top = 7/3$ and $(0, 7) \cdot (1/3, 2/3)^\top = 14/3$. The other cases are verified similarly. $\square$

The constructed dataset has $N = 4d + m + 1$ samples in dimension $n = 2d + 1$. This completes the polynomial-time many-one reduction proving Theorem ?? in the case of $d_K = d_V = 1, p_t \leq 2$.

4 Hardmax exact-fit is NP-hard and NP-complete at $d_K = 1$

We next turn to hardmax attention. NP-hardness holds for all fixed key and value dimensions (Theorem ??). When the key dimension is one, an additional NP-membership argument upgrades this to NP-completeness (Theorem ??).

Theorem 2 (Hardmax attention is NP-hard). *For every fixed $d_K \geq 1$ and $d_V \geq 1$, HARDMAX-EXACT-FIT is NP-hard. Hardness already holds when every input sequence has length $p_t \leq 2$ and every entry of every X_t is Boolean (in $\{0, 1\}$).*

Theorem 3 (Hardmax attention is NP-complete). *For every fixed $d_V \geq 1$ and $d_K = 1$, HARDMAX-EXACT-FIT is in NP, and hence (by Theorem ?? at $d_K = 1$) is NP-complete.*

4.1 Proof of Theorem ??

The hardmax reduction keeps the Boolean core of the softmax construction but removes the auxiliary vector F : truth is encoded directly by values 0 and 1. Given a distinct-literal 1-IN-3-SAT instance ϕ on d variables $x_1, \dots, x_d$ and m clauses c_j (Lemma ??), set $n = 2d$ and use the coordinate basis vectors

$$L_1, \dots, L_d, \bar{L}_1, \dots, \bar{L}_d \in \{0, 1\}^n.$$

For each variable i and clause j , define the vectors

$$P_i = L_i + \bar{L}_i, \quad D_j = T(\ell_{j,1}) + T(\ell_{j,2}) + T(\ell_{j,3}).$$

By Lemma ??, D_j is a sum of three distinct basis vectors. Every input vector therefore has Boolean entries in $\{0, 1\}$.

The dataset $\mathcal{D}_\phi$ consists of $2d + m$ samples in three groups, with $d_v = 1$:

Group	Sample X_t	Target Y_t	Count
1	$[P_i]$	1	d
2	$[L_i, \bar{L}_i]$	$(1, 0)$	d
3	$[D_j]$	1	m

Soundness. Suppose Θ exactly fits $\mathcal{D}_\phi$.

Group 1. The sample $[P_i] \mapsto 1$ has trivial hardmax weight 1, so $v(P_i) = 1$. By linearity of W_v ,

$$v(L_i) + v(\bar{L}_i) = 1. \quad (5)$$

Group 2. For the length-two sample $[L_i, \bar{L}_i] \mapsto (1, 0)$, each of the two output columns equals a hardmax-weighted average of $v(L_i)$ and $v(\bar{L}_i)$. If column j of the score matrix has both entries equal, the equal-weight tiebreak rule (Section ??) assigns weight $1/2$ to each, giving column- j output $\frac{1}{2}(v(L_i) + v(\bar{L}_i)) = \frac{1}{2}$ by (??), contradicting both targets 1 and 0. Hence both columns have a strict winner: column 1 selects either L_i or $\bar{L}_i$, forcing the selected value to equal 1; column 2 analogously forces the selected value to equal 0. Together with (??),

$$\{v(L_i), v(\bar{L}_i)\} = \{0, 1\}.$$

Define the Boolean assignment $x_i = \text{TRUE}$ iff $v(L_i) = 1$, so a literal ℓ is true iff $v(T(\ell)) = 1$.

Group 3. The length-1 sample $[D_j] \mapsto 1$ gives $v(D_j) = \sum_{r=1}^3 v(T(\ell_{j,r})) = 1$. Each summand is in $\{0, 1\}$, so exactly one of the three literals of c_j is true.

Completeness. If ϕ is satisfiable, fix a satisfying assignment and define row vectors $W_v, W_Q \in \mathbb{R}^{1 \times n}$ by

$$W_v T(\ell) = W_Q T(\ell) = \begin{cases} 1, & \ell \text{ is true,} \\ 0, & \ell \text{ is false,} \end{cases}$$

and $W_K \in \mathbb{R}^{1 \times n}$ by $W_K L_i = 1, W_K \bar{L}_i = -1$ for every i (independent of truth). Group 1 holds because exactly one of $L_i, \bar{L}_i$ is true, so $v(P_i) = 1 + 0 = 1$. Group 3 holds because each clause has exactly one true literal, so $v(D_j) = 1 + 0 + 0 = 1$.

For Group 2 when x_i is true, $W_Q L_i = 1, W_Q \bar{L}_i = 0$; the score matrix of $[L_i, \bar{L}_i]$ is $\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$. Column 1 has unique max in row 1 (since $1 > 0$), giving weights $(1, 0)^\top$ and output $v(L_i) = 1$. Column 2 has unique max in row 2 (since $0 > -1$), giving weights $(0, 1)^\top$ and output $v(\bar{L}_i) = 0$. The output is $(1, 0)$, matching the target. The case when x_i is false is similar.

4.2 The $d_K > 1$ and $d_v > 1$ case

The reductions of Sections ?? and ?? are constructed at $d_K = d_v = 1$ and $p_t \leq 2$; the extension to general d_K, d_v below applies to both Theorems ?? and ??.

General sequence length. For every fixed $k \geq 2$, the problem restricted to $p_t \leq k$ inherits NP-hardness from the $p_t \leq 2$ case.

Parameter dimension. For arbitrary fixed $d_K > 1$, replace the row-vector parameters $W_Q, W_K \in \mathbb{R}^{1 \times n}$ of Lemma ?? (resp. of Section ??) by matrices $\bar{W}_Q, \bar{W}_K \in \mathbb{R}^{d_K \times n}$ obtained by placing W_Q (resp. W_K) in the first row and padding zeros in the remaining $d_K - 1$ rows. Then the NP-hardness results in Theorems ?? and ?? extend to every fixed $d_K \geq 1$.

Dimension of samples. For arbitrary fixed $d_v > 1$, replicate every scalar target $Y_t \in \mathbb{Q}^{1 \times p_t}$ into a $d_v \times p_t$ matrix $\bar{Y}_t$ whose every row equals Y_t , and replace $W_v \in \mathbb{R}^{1 \times n}$ by the matrix $\bar{W}_v \in \mathbb{R}^{d_v \times n}$ whose every row equals W_v . Then the NP-hardness results in Theorems ?? and ?? extend to every fixed $d_v \geq 1$.

4.3 NP-completeness at $d_K = 1$ (proof of Theorem ??)

Theorem ?? at $d_K = 1$ already gives NP-hardness. To obtain NP-completeness it suffices to show NP-membership, which we do by exhibiting a polynomial-bit-length rational certificate.

Lemma 5 (Rational certificate for linear feasibility). *A consistent finite system of rational linear equations and weak/strict inequalities in ν real variables, whose constraint coefficients have total bit length B , has a rational solution of bit length polynomial in $\nu + B$.*

This is standard rational linear-programming bit-complexity; mixed weak/strict rational systems are handled by the well-known reduction to standard LP feasibility ?.

Proof of NP-completeness in Theorem ??. It remains to encode the exact-fit constraints as rational linear (in)equalities.

For each sample $X_t \in \mathbb{Q}^{n \times p_t}$ at $d_K = 1$, the score matrix factorizes as

$$X_t^\top W_Q^\top W_K X_t = (W_Q X_t)^\top (W_K X_t) \in \mathbb{R}^{p_t \times p_t},$$

a matrix of rank at most one whose i th column is $(W_K X_t)_i \cdot W_Q X_t$. The hardmax pattern of column i , i.e. the set of coordinates achieving the column maximum (and hence receiving equal hardmax weight $1/k$, with k the size of that set), depends only on the sign of $(W_K X_t)_i$ and on the argmax (resp. argmin) of $W_Q X_t$:

- If $(W_K X_t)_i > 0$: column i of $\text{hardmax}(X_t^\top W_Q^\top W_K X_t)$ assigns weight $1/k$ to each of the k coordinates achieving $\max_j (W_Q X_t)_j$, and 0 elsewhere.
- If $(W_K X_t)_i < 0$: same with argmin in place of argmax (and the k coordinates achieving $\min_j (W_Q X_t)_j$).
- If $(W_K X_t)_i = 0$: column i is uniform with weight $1/p_t$ in every coordinate.

Thus a certificate only needs to record a finite sign-and-order pattern for each column.

In other words, we can record (i) the sign of $(W_K X_t)_i$, and (ii) the equality/inequality pattern among the coordinates of $W_Q X_t$ that specifies the maximum and minimum. Once those patterns are known, $\text{hardmax}(X_t^\top W_Q^\top W_K X_t)$ is determined. For example, if $(W_K X_t)_i > 0$, and the first and second coordinates of $W_Q X_t$ achieve maximum, then W_Q satisfies $(W_Q X_t)_1 = (W_Q X_t)_2, (W_Q X_t)_1 > (W_Q X_t)_3, \dots, (W_Q X_t)_1 > (W_Q X_t)_{p_t}$. Furthermore, for any W_Q, W_K satisfying the above conditions, the i th column of $\text{hardmax}(X_t^\top W_Q^\top W_K X_t)$ must be $(\frac{1}{2}, \frac{1}{2}, 0, \dots, 0)$.

Once the pattern is fixed, the matrix $H_t := \text{hardmax}(X_t^\top W_Q^\top W_K X_t)$ becomes a constant rational matrix, and the exact-fit equation $W_V X_t H_t = Y_t$ is rational-linear in W_V . Aggregating across all t , we obtain a finite rational system of linear equations and inequalities in Θ , satisfied by the original real solution by construction. By Lemma ??, there exist rational parameters Θ of polynomial bit length that satisfy the system of equations and inequalities, and solves HARDMAX-EXACT-FIT. $\square$

The argument relies on $d_K = 1$ for the rank-one factorization. For $d_K \geq 2$, NP-completeness is left open.

5 Concluding remarks

One way to demonstrate that a problem is hard is to show that even the simplest case is already hard. Blum and Rivest did this for neural-network training ?; here we do it for a single attention head. Our results concern *exact* fitting of a single attention head in the worst case. For softmax we prove only NP-hardness; membership in NP is left open because the score matrix involves transcendental exponentials of real-valued inner products. This leaves several open problems, which we mention below.

Beyond NP for softmax. Where does softmax exact-fit sit within the hierarchy of real-feasibility complexity classes? Standard $\exists\mathbb{R}$ -completeness in the algebraic sense of Abrahamsen, Kleist, and Miltzow ?, and of Bertschinger, Hertrich, Jungeblut, Miltzow, and Weber ?, does not apply directly

here because the softmax constraints involve transcendental exponentials rather than polynomial relations; the natural target is $\exists\mathbb{R}_{\text{exp}}$, the existential theory of the reals with exponentiation.

Inapproximability. Hardness of approximation under ℓ_2 error (rather than ℓ_∞) is a natural further direction; this likely requires invoking standard inapproximability tools (e.g., ??).

Polynomial-time algorithms. A complementary direction is to identify special classes of attention or transformers with polynomial-time algorithms, perhaps with runtime polynomial in the number of samples while remaining exponential in the dimensions, as in ?.

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